



Tecnológico de Monterrey
Partial Exam 2 A
2026-Jan-May

Grade

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Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 24/03/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Multiple option section. Read the following questions and identify the option that best answers each, then write its letter on the line.

1. (4 points) _____ Identify the quotient rule for derivatives.

(a) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{u}{v}\right)' = \frac{v'u - u'v}{u^2}$ (d) $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{u^2}$.

2. (4 points) _____ Identify the product rule for derivatives

(a) $(uv)' = u'v + uv'$ (b) $(uv)' = u'v + u'v$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u' + v'$

3. (4 points) _____ Identify the chain rule for derivatives

(a) $[u(v(x))]' = v'(u(x))$ (b) $[u(v(x))]' = u'(v(x))$ (c) $[u(v(x))]' = u'(v(x))v'(x)$ (d) $[u(v(x))]' = v'(u(x))u'(x)$

4. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}n^x = xn^x$ (b) $\frac{d}{dx}n^x = n^x \ln(n)$ (c) $\frac{d}{dx}n^x = ne^{x-1}$ (d) $\frac{d}{dx}n^x = \frac{1}{x}n^x$

5. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \cos(x) = \cos(x)$ (b) $\frac{d}{dx} \cos(x) = -\sin(x)$ (c) $\frac{d}{dx} \cos(x) = \sin(x)$ (d) $\frac{d}{dx} \cos(x) = -\cos(x)$

6. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \sin(x) = \cos(x)$ (b) $\frac{d}{dx} \sin(x) = -\sin(x)$ (c) $\frac{d}{dx} \sin(x) = \sin(x)$ (d) $\frac{d}{dx} \sin(x) = -\cos(x)$

7. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \log_n(x) = \frac{1}{\ln(x)}$ (b) $\frac{d}{dx} \log_n(x) = \frac{1}{x \ln(n)}$ (c) $\frac{d}{dx} \log_n(x) = \frac{\ln(x)}{x}$ (d) $\frac{d}{dx} \log_n(x) = \frac{1}{x \ln(x)}$

II Derivatives by rules. Use the rules of derivatives that can be usefull to compute the derivatives of the following functions.

- 1.** (10 **points**) Compute the derivative of the function $f(x) = e^{-x^2} \sin^2(x)$.

- 2.** (10 **points**) Compute the derivative of the function $g(x) = \log [xe^{-x}]$.

III Critical points. In this section your task is to find all critical points of the functions in the given context.

1. (13 **points**) An open box is made from a 5×5 cm sheet by cutting squares of side x from the corner. The volume of the box is given by $V(x) = x(5 - 2x)^2$. Find the length of the squares that achieves the greater volume.

2. (13 **points**) The displacement of the planets over time can be modeled with the following function, $s(t) = 10 \sin\left(\frac{\pi}{5}t\right) + 5 \cos\left(\frac{\pi}{5}t\right)$. Find the time at which the planets are in a maximum and minimum displacements.

3. (13 points) The concentration of a drug in blood after t hours is given by $C(t) = te^{-kt}$. Find the time that gives the maximum concentration in terms of k .

4. (13 points) A rectangular plot with one side along a river needs fencing on three sides. If the area is fixed A and the cost is given by $C(x) = px + 2p\frac{A}{x}$, where x is the side along the river. Compute side length of the side of the river that minimize the fencing cost given that $A = 22500$ and $p = 100$.